

Funding a Venture

Prof. Dr. Orestis Terzidis



INTRODUCTION

Key Activities

- Entrepreneurs identify opportunities, mobilize resources, execute on their vision and manage risks.

Inspired by Byers et al. 2011, Pos. 497

- Entrepreneurship is focused on the creation of a new enterprise that serves society and makes a positive change.

Inspired by Byers et al. 2011, Pos. 506



Labor Market



Customer Market



Value
Creation
System

Value
Delivery
System

Value
Capture
System



Supplier Market

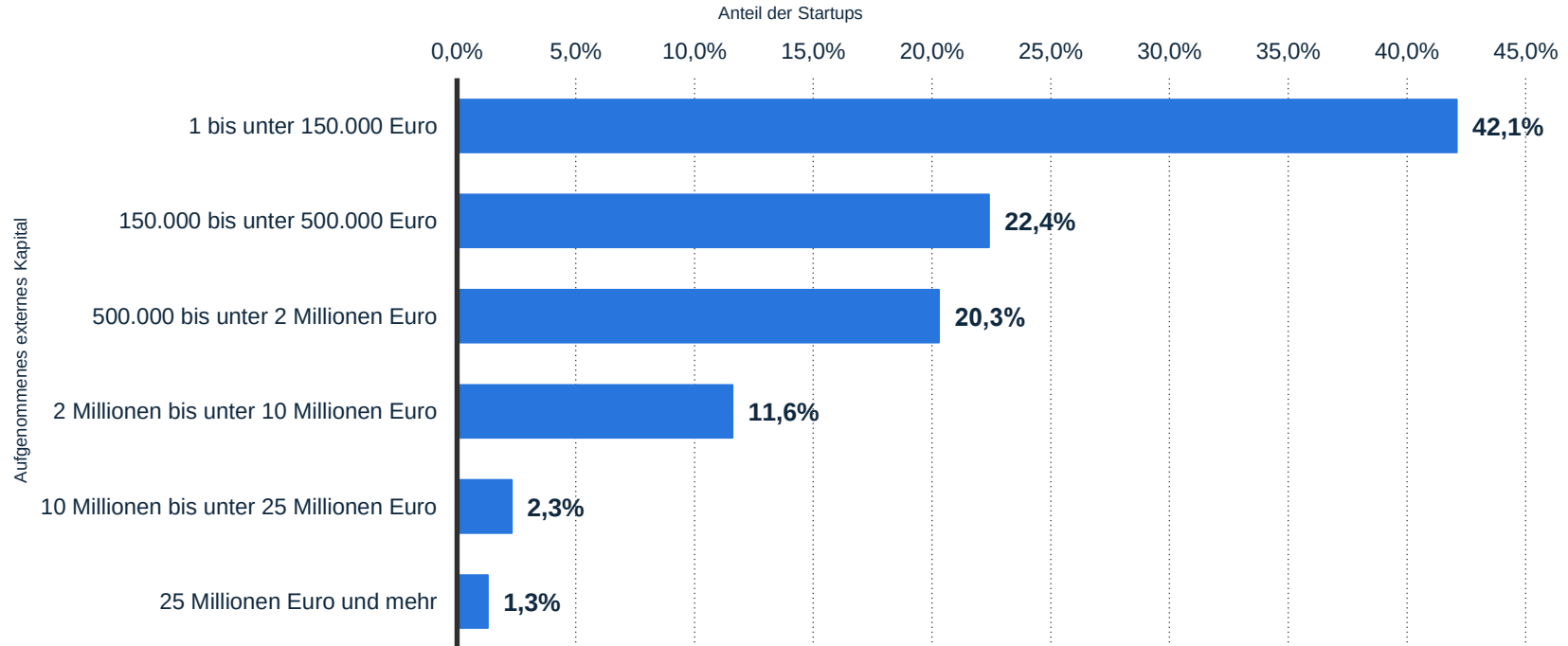


Partner Market



Financial Market

External capital raised by start-ups in Germany until 2019

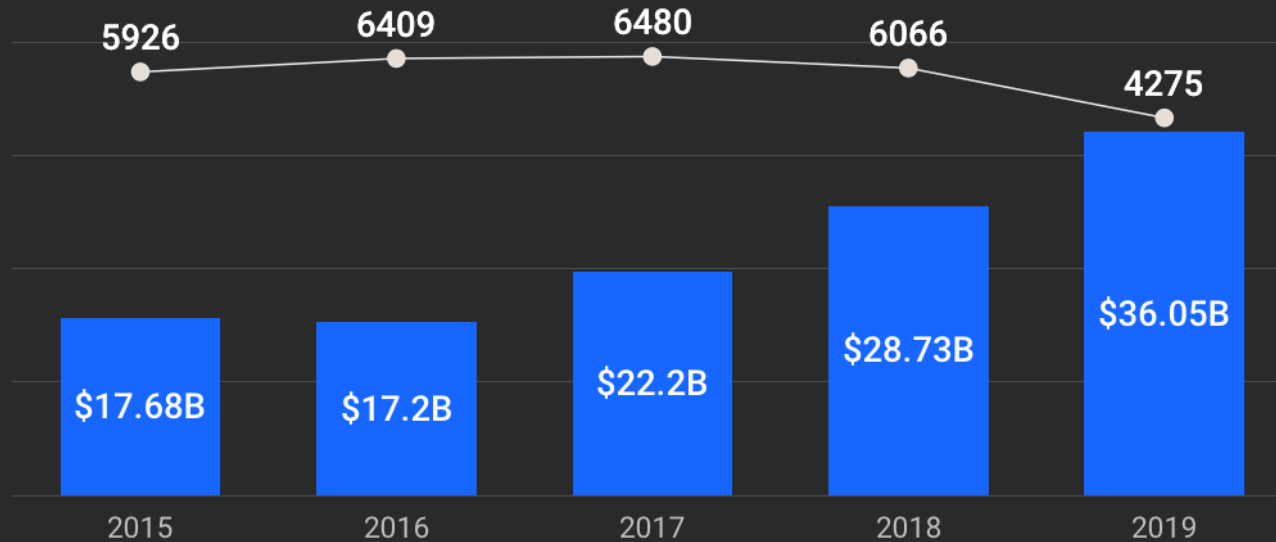


European Startups Annual Deal & Dollar Volume

Data is current through early January 2020 and is subject to change as new transactions get added to Crunchbase over time. Dollar volume based on deals of known size; round counts are for all deals. Foreign currencies were converted to USD.

■ Total \$ Invested

● Number of Deals



crunchbase news

Learning objectives

Learn about the possibilities of funding a venture

- What are typical sources of capital?

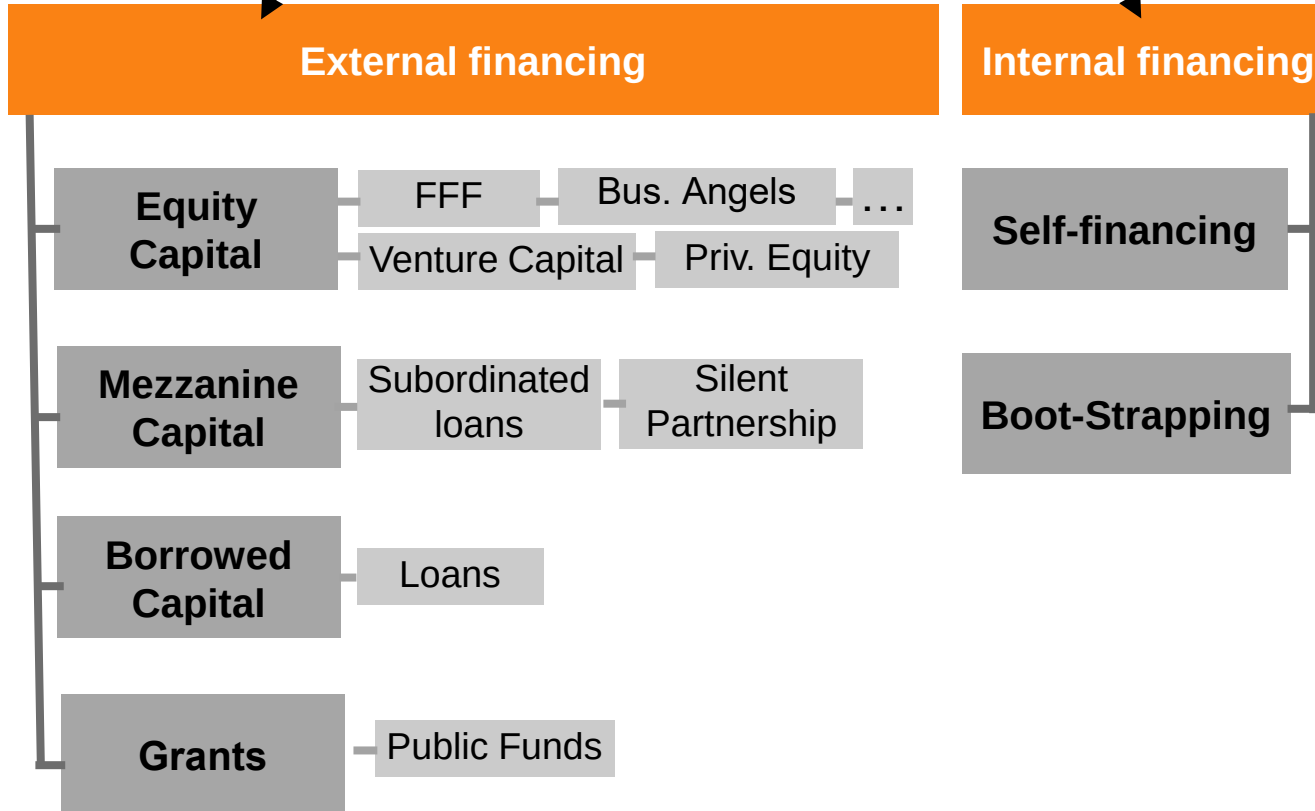
Learn to choose the appropriate source of capital

- Align the stages of the start-up life cycle with the start-up financing life cycle.
- Know the characteristics of the different sources of funding.

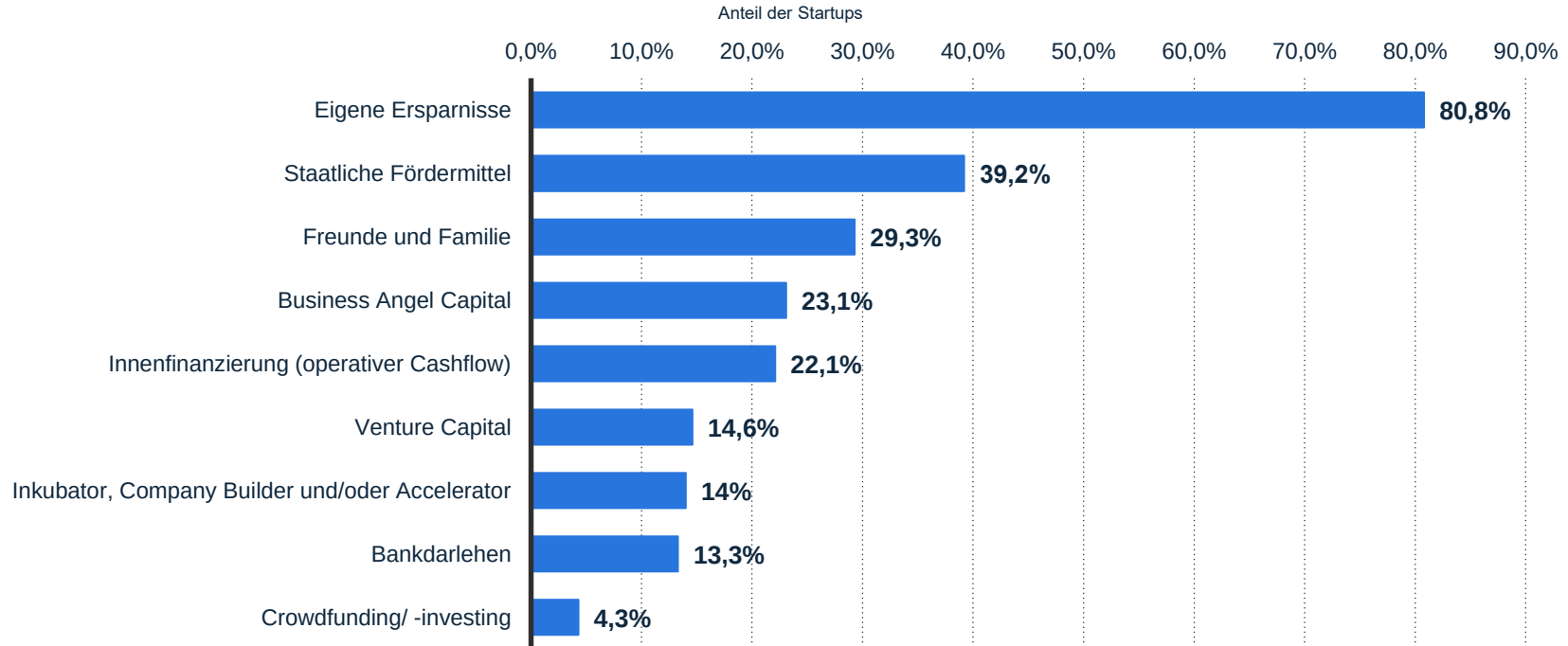
Understand investment risk

- What is the risk of the investors?
- What are your legal obligations in case of a company crisis.

Funding a Start-up



Sources of Funding of Startups in Germany



External Financing

Equity capital

- Having a share implies to co-own the company's
- Control rights, co-determination rights, information rights
- Direct participation in business success & failure

Mezzanine capital

- Repayment obligation, profit-linked interest
- Optional control rights, co-determination/information rights
- Potential participation in failure

Loans/Borrowed capital

- Repayment obligation, interest payment
- Investors do not hold any shares
- No participation in business success

UNDERSTANDING RISK: INSOLVENCY

A **risk** is an indication of the **possibility that** with some probability **a loss may occur** in connection with a **decision** or an expected benefit may not materialize.

Crisis and Bankruptcy

Legal definition:

A company is in a **crisis**, if it does not obtain loans at market prices anymore. It is then said to be '**unworthy of credit**'.

Bankruptcy (or insolvency) is a **legal status of a firm** that cannot repay the debts it owes to creditors.

- Bankruptcy is imposed by a **court order**.
- The court usually appoints a **liquidator** (Insolvenzverwalter).
- In many cases, the goal of bankruptcy is to **liquidate company assets** and 'fairly' **satisfy the creditor claims**.

Insolvency code (Insolvenzordnung InsO)

Reasons for insolvency (InsO §17-19)

- Illiquidity (Zahlungsunfähigkeit)
- Imminent insolvency (Drohende Zahlungsunfähigkeit)
- Over-indebtedness (Überschuldung)

For a company with limited liability (GmbH), as an owner in case of insolvency you only lose the nominal capital. Your private current and future wealth is delimited.

Liability of the management

- **Delayed filing** for insolvency (Insolvenzverschleppung): 3 weeks
- If the company you manage enters a crisis, you must act and inform the authorities.



Example

Creditors

Bank loan	60 k€
Mezzanine	40 k€
Salaries	20 k€
Supplier	50 k€
Supplier 2	30 k€
Social Security	15 k€
Tax payments	10 k€
Sum	225 k€

(Potential) Revenue

Sales contract A	150 k€
Mature Lead B	200 k€
New Lead C	100 k€
Sum	350 k€

Example: Cancellation of Mature Lead

Creditors

Bank loan	60 k€
Mezzanine	40 k€
Salaries	20 k€
Supplier 1	50 k€
Supplier 2	30 k€
Social Security	15 k€
Tax payments	10 k€
Sum	225 k€

(Potential) Revenue

Sales contract A	150 k€
Mature Lead B	200 k€
New Lead C	100 k€
Sum	150 k€

Risk classes

The risk of your investors depends on the type of the investment.

Equity capital

- Highest liability risk
- Lowest rank in case of insolvency or liquidation

Mezzanine Capital

- Lower ranked than borrowed capital
- Higher interest rate than loans

Loans/borrowed capital

- Reduced risk
- Highest rank in case of insolvency or liquidation



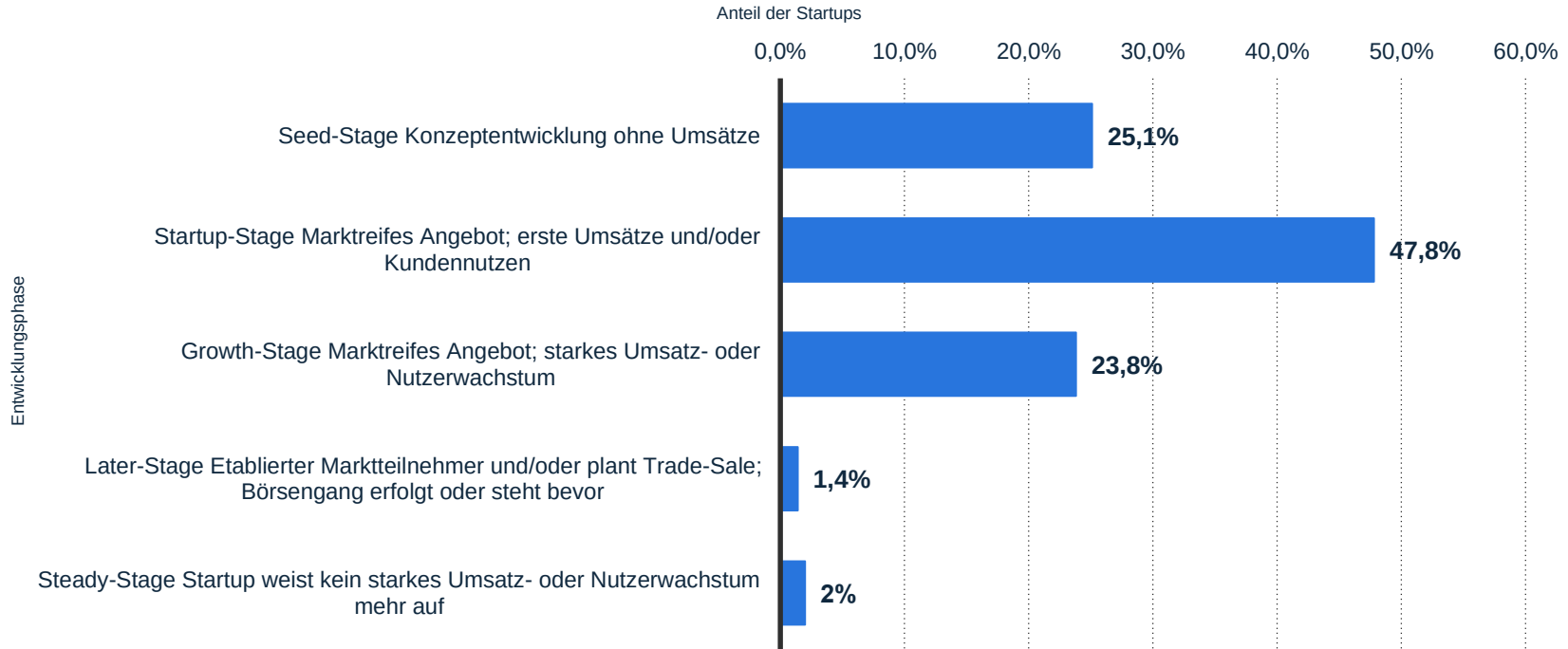
START-UP FINANCING CYCLE

Start-up Lifecycle

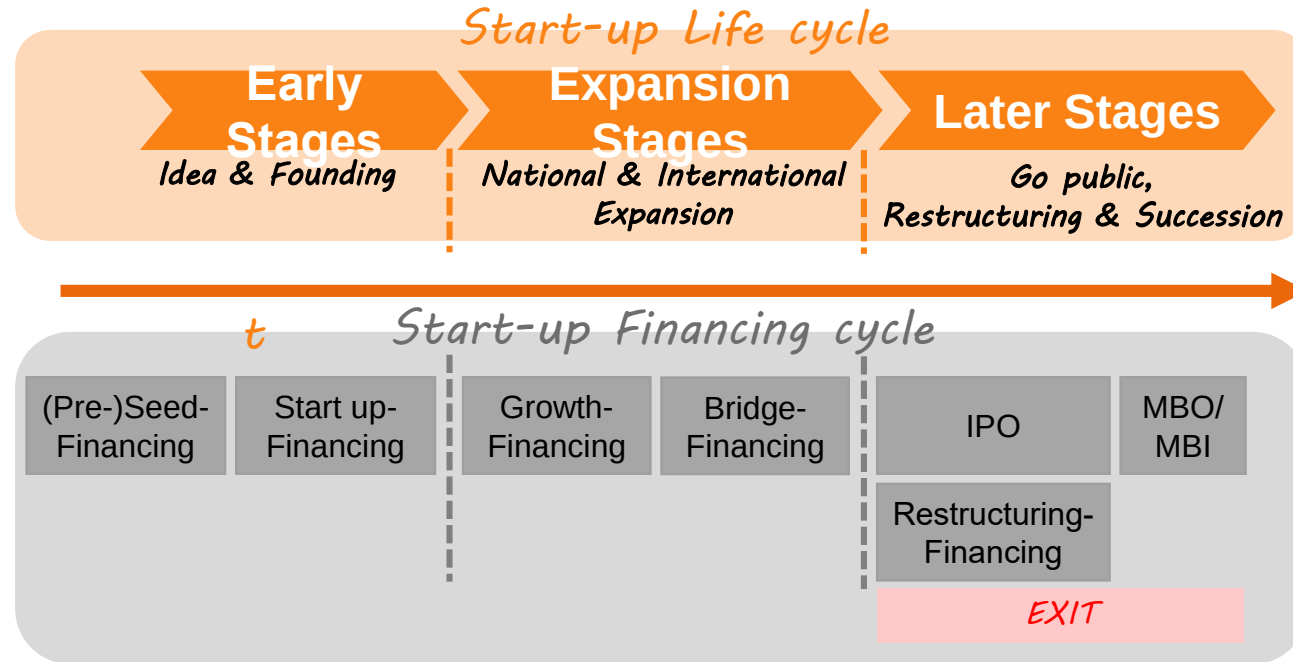
Startups go through typical phases. The focus and the context is different in different phases.



Distribution of Start-ups in Germany 2019

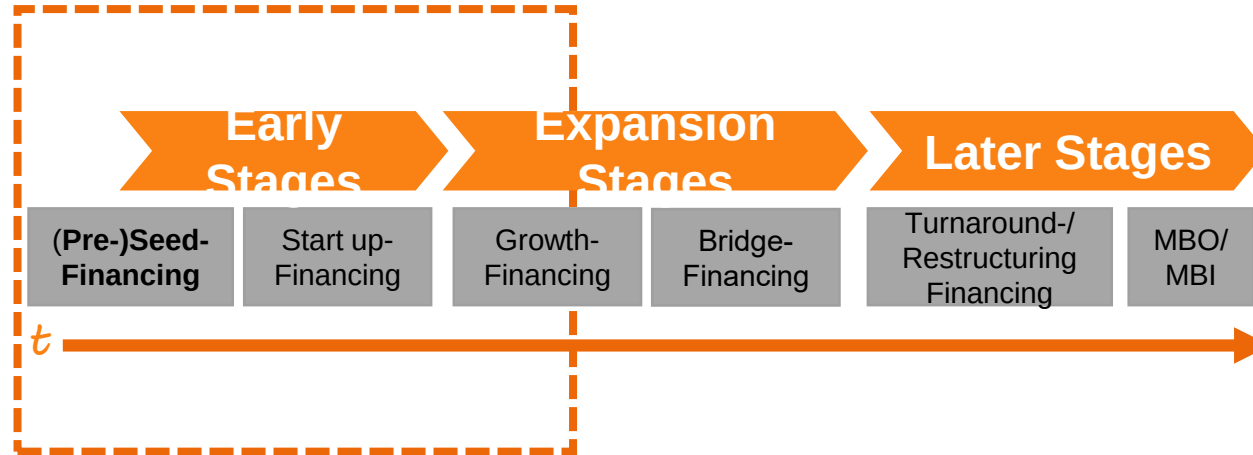


Start-up Financing Cycle



The different stages of the start-up life cycle are characterized by different stages of funding.

Start-up Financing Cycle



We will focus on the early stages in the start-up life cycle.

Early Stages: Idea phase/(Pre)-Seed phase

Why is it difficult for start-ups to obtain loans?

- Lack of securities
- No track record
- Irreversibility of R&D costs (sunk cost)
- Risk of failure
- High uncertainty about
 - Market opportunities
 - Development of the company
- Asymmetric information
 - Skills of the founders? → no company history
 - Difficulty to evaluate the quality of the product/technology



Credit rationing: limitation of credit amount or no credit

Early Stages: Idea phase/(Pre)-Seed phase

Characteristics of this phase :

- Idea generation, prototyping, feasibility studies, team building etc.
- No revenues, no profits/moderate losses
- Financing from own cash-flows is not possible

Typical sources of seed funding:

- Own Equity capital, 'Fools, Friends and Family'
- Business Angels
- Crowd Funding
- Incubators
- Public Grants: e.g. Exist Business Start-up Grant
- Seed investors: e.g. High Tech Gründerfonds (HTGF)



Early Stages: Start-up Phase

Characteristics of this stage:

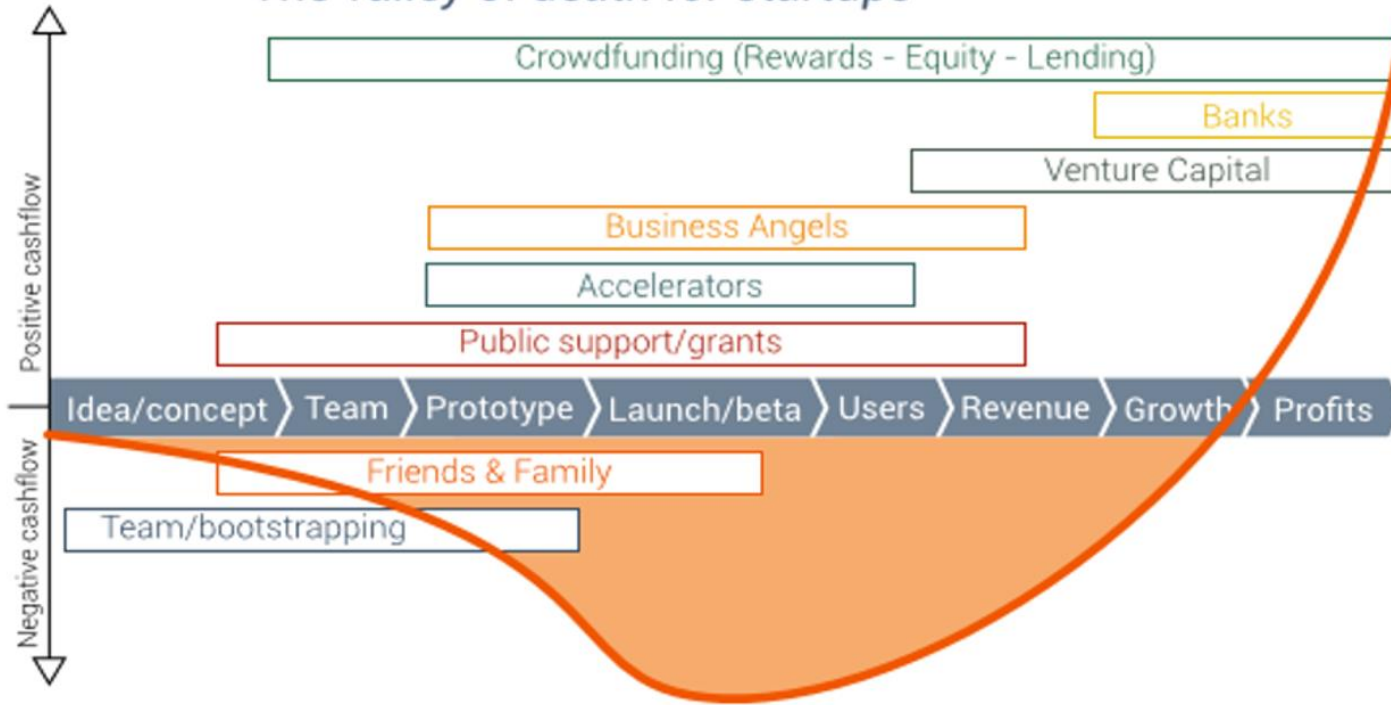
- Company foundation, product development reaches production stage, first marketing concepts, partnerships
- First revenues, first small profits/high losses
- Rising capital requirements

Sources of financing:

- Equity capital
- Business Angels
- Crowd Funding
- Public Funds
- Venture Capital



The valley of death for startups



BUSINESS ANGELS

Investor's View: The Hippodrome Metaphor

There is no question that irrespective of the **horse (product)**, **horse race (market)**, or **odds (financial criteria)**, it is the **jockey (entrepreneur/s)** who fundamentally determines whether the venture capitalist will place a bet at all.

MacMillan, Siegel and Narasimha 1985)



Business Angels

- BA are wealthy private investors, which typically have their own business.
- They provide their capital, but also their know-how and experiences as well as their personal network (social capital).
- In exchange they acquire shares of the company. Typically, after 4 - 7 years they sell their shares to realize profits (or losses).
- Typical investment sum: 25,000 Euro – 500,000 Euro



Business Angels' Investment Criteria

- Empirical study in cooperation with a local business angel (Dr. Friedrich Georg Höpfner)
- Investment Criteria of German Business Angels

4 categories:

- Entrepreneur & Management Team
- Market & Product
- Financial Factors
- Other Business Attributes



Characteristics Business Angels

Age (mean):	48
Gender:	96% male
Annual income:	125 - 250 T EUR
Overall wealth:	2,5 - 5 Mio. EUR
Academic qualification:	96% Master or PhD
Entrepreneurial experience:	83% Self-employed 73% Consultant/Coach
Professional background:	98% business - 44% technical
Ø Capital:	25,000 EUR – 500,000 EUR

Motivation of Business Angels for Investing

	Mean Ranking	SD
Supporting young entrepreneurs	2,31	0.679
Contribute own professional experience	2,31	0,788
For fun	2,32	0,900
Potentially fruitful investment	2,46	0,706
To play a role in the entrepreneurial process	2,85	0,881
Support socially beneficial products	3,15	0,881
Regional support	3,19	1,132
Positive recognition in community	3,92	0,796
Help friends or family	4,00	0,938
Tax purposes	4,12	0,781

Note: Respondents ranked variables:

1 = extremely important; 2 = very important; 3 = important; 4 = indifferent; 5 = totally unimportant

Top Factors per Category

Entrepreneur & team

- Trustworthiness
- Enthusiasm
- Achievement motivation
- Ability to communicate the product
- Frustration tolerance

Financial Factors

- Proportion of self-financing by entrepreneur
- Anticipated financial rewards

Market & Product

- Product's sales potential
- Growth potential of market
- Product meets pledged properties
- Unique feature of product

Other Business Attributes

- Investor's involvement possible (contribute skills)

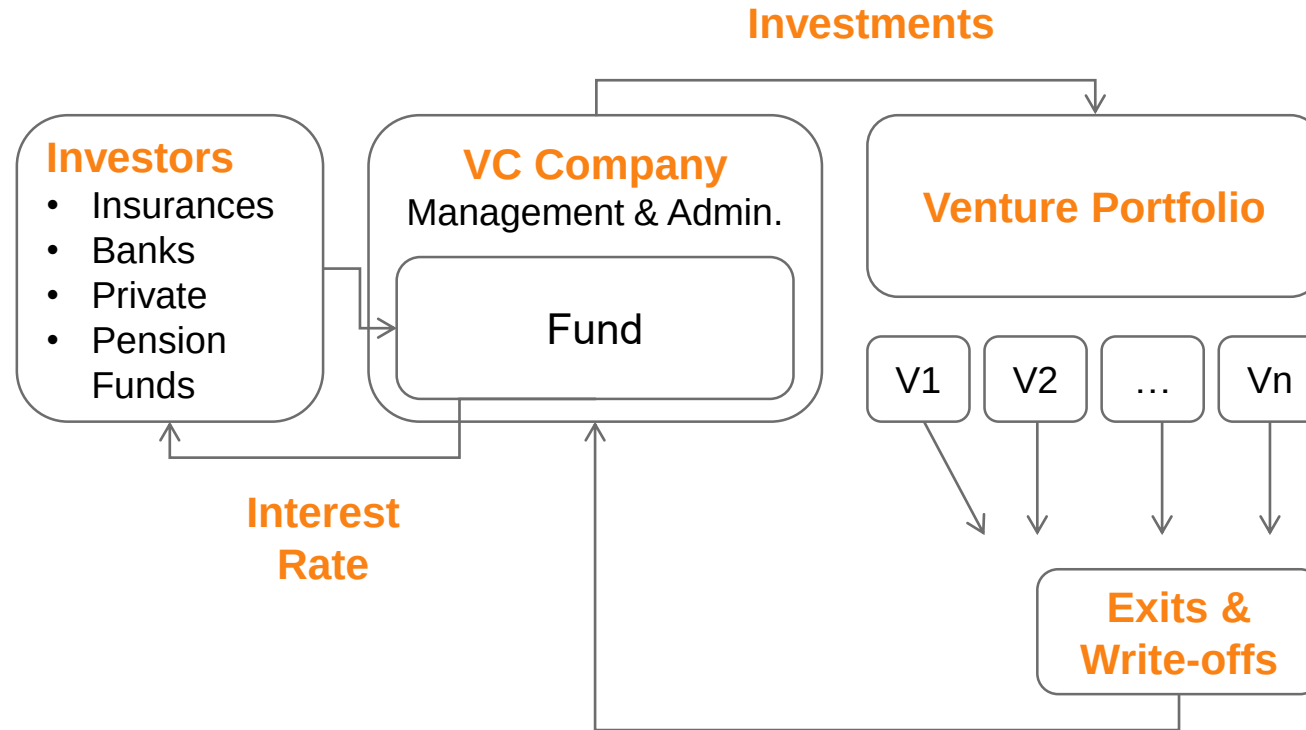
VENTURE CAPITAL

Venture Capital

- VC investors provide equity capital for start-ups.
- They share the risk with the entrepreneur(s).
- There is no interest and repayment.
- VCs realize profits through selling their equity after some years (exit). Typical period: 3-7 years.
- VCs have controlling and voting rights.
- The key objective of a VC is to increase the value of the start-up. VCs only invest if they anticipate exit opportunities
- Typical phases of the start-up financing cycle are start-up financing and growth financing.
- Be prepared to shoot high.



Structure of VC Business



The Six Phases of the Funding Process

- Deal Origination
- Screening
- Evaluation
- Deal Closing
- Post-Investment Activities
- Exit



Deal Origination

The investor's goal is to identify new companies.

- Through direct contact from the entrepreneur (e.g., cold call, email, application platform).
- Through an active search for deals.
- Through a referral process.

Referral process: this is one of the reasons why **building social capital** (networking) is very important.



Screening

- The VC evaluates the identified ventures considering a limited number of exclusion criteria.
- An unacceptable value for one criterion cannot be compensated by a high value of another one.
- Given the large number of proposals collected, the objective is to efficiently create a shortlist of ventures. Most proposals are rejected quickly.
- Check generic aspects: deal size, industry, technologies, geographic location, stage of financing.
- Check specific aspects: The VC performs a first assessment of the selected business plans.



Evaluation

- The investor performs an in-depth analysis of the company.
- The business plan is analyzed in detail and other sources of information are considered.
- Interviews with all members of the team,
- Market value of similar companies,
- Take a tour of the venture facilities,
- Request an expertise,
- Ask for information about customers.

A low but acceptable value on one assessment criterion can be compensated by a high value on another.



Deal Closing

Once the investor has decided to invest in the company, s/he proceeds with the closing of the deal.

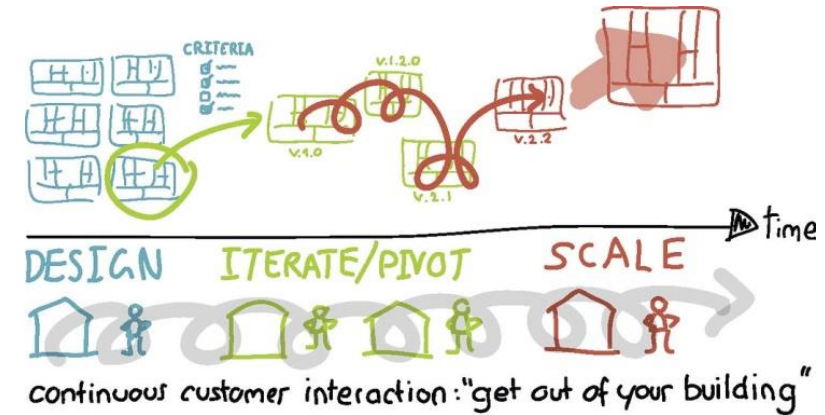
At this point, the terms of the investment are negotiated:

- Price and equity stake are defined,
- Legal aspects are considered,
- Milestones, payment conditions and exit clauses are defined.
- Typically the payout is not done in a single step, but depends on the attainment of agreed goals.



Post-investment Activities

- To protect the investment and increase the probability of success, many investors provide a series of value-added activities over time.
- The approach to the support provided usually varies according to the type of investor.
- The quality of the investor, his network and expertise, may be crucial for the success of the venture. It is about “smart money”.



Exit

- When the company value has reached a sufficient level, or the planned duration of the investment is reached, the investor proceeds with the exit.
- An investor can make an exit through several events (Cumming and MacIntosh 2003):
- Acquisition: the company as a whole is sold.
- Initial public offering (IPO): the investor sells the shares on the stock market.
- Buy-back: the investor sells the shares to the founders.
- Secondary sale: the investor sells shares to another investor.
- Write off: the investor realizes a capital loss.



If failure is defined as **failing to see the projected return on investment** - say, a specific revenue growth rate or date to break even on cash flow - then **more than 95% of start-ups fail**.

Shikhar Ghosh (Harvard Business School)

Gage, D. (2012) "The venture capital secret: 3 out of 4 start-ups fail", [online], The Wall Street Journal U.S. Edition, updated Sept. 20, 2012

Top Reasons why Startups Fail

No Market Need

Run out of Cash

Not the Right Team

Got Outcompeted

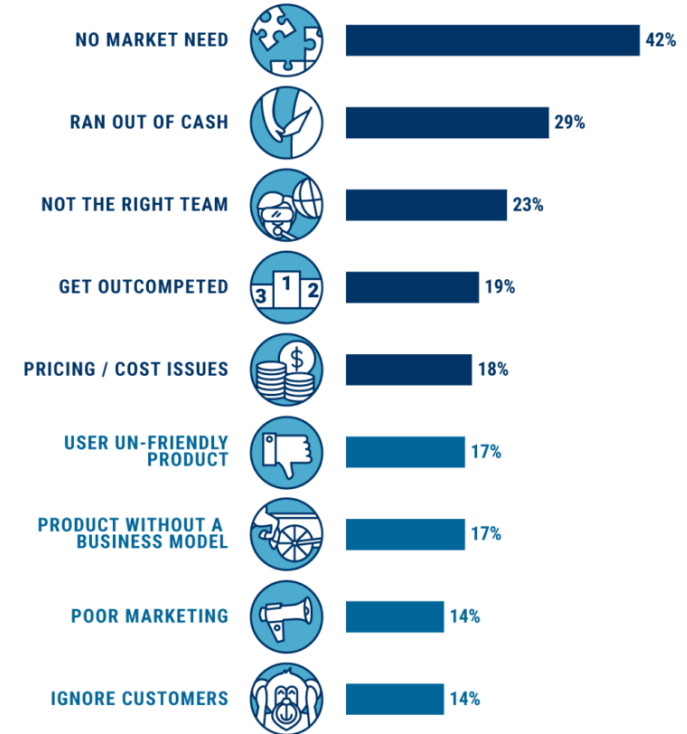
Pricing/Cost Issues

User Un-friendly Product

Product without a business model

Poor Marketing

Ignore Customers



EXAMPLE HTGF INVESTMENT

High Tech Gründerfonds (HTGF)

- The HTGF is a public-private venture capital investment firm which has about €886 million under management in three funds.
- It is an early stage seed investor, focused on high potential high-tech start-ups.
- The seed financing is provided to allow start-ups to take their ideas through the prototyping phase up to the market launch.
- Usually High-Tech Gründerfonds invests about €600,000 in the seed stage and up to a total of €3 million per portfolio company in later rounds.



Investment Example / Terms

- HTGF buys 5,000 shares at nominal value (1 EUR/share) to own 15 % of shares/equity.
- 595,000 EUR are given as unsecured subordinated convertible loan without discount.
- Interest rate of 6% p.a. payment deferred during the first 4 years.
- No company evaluation is done.
- Additional subordinated shareholder loan with conversion rights.
(Gesellschafterdarlehen mit Rangrücktritt und Wandlungsrecht).
- Conversion of loan and interests into equity in the case of further investment rounds.

(Simplified) Example: 1st Round at t1

Partner	Nominal Capital [€]	Agio [€]	Sum [€]	Shares [%]	Comment
Founders	25,000	75,000	100,000	75	
Side Investor	3,350	196,650	200,000	10	Evaluation after negotiation
HTGF	5,000	595,000	600,000	15	Agio is a subordinate, convertible loan
Sum	33,350	866,650	900,000	100	

- An overall of 900,000 € are available for the venture (equity + agio).
- Only 33,350 € are nominal capital registered at the notary.
- Assume only multiples of 50 € in the nominal capital.
- There is no common evaluation.

Note: In the exam we use the English number format! (→ 1000 € = 1,000 €)!

2nd Round at t2

- A capital increase is done to allow the firm to evolve.
- After negotiations and estimations of external experts, the **pre-money evaluation** of the company is considered to be **6 Mio €**.
- The new investor wants to invest **1.7 Mio €**.
- **HTGF** decides to convert part of the loan and the interest in order **not be diluted**.

Questions:

1. How much money must the HTGF convert?
2. What is the post-money evaluation?
3. How is the nominal capital increased to reflect the investment?
4. What are the (rounded) nominal shares?

1. How much money must the HTGF convert?

The post-money evaluation (POM) is given by

$$\text{POM} = 6 \text{ Mio €} + 1.7 \text{ Mio €} + \text{CON}$$

where CON is the sum converted by HTGF.

If the pre-money evaluation of the venture is 6 Mio €, and HTGF owns 15% shares, this corresponds to 900,000 €. With these facts, CON is given by the relation:

$$(900,000 \text{ €} + \text{CON}) / (7,700,000 \text{ €} + \text{CON}) = 15\%$$

$$\text{Then: CON} = 1/0,85 * (0,15 * 7,700,000 \text{ €} - 900,000 \text{ €}) = 300,000 \text{ €}$$

Answer: **The HTGF must convert 300,000 € of the loan.**

2. What is the Post-Money Evaluation?

The post-money evaluation is

$$\text{POM} = 6,000,000 \text{ €} + 1,700,000 \text{ €} + 300,000 \text{ €}$$

Answer: **POM = 8,000,000 €**

3. How is the nominal capital increased?

Nominal capital before the second round: 33,350 €

Corresponds to the value of 6 Mio €.

New value of 8 Mio €.

Therefore, the new nominal capital NNC is

$$\text{NNC} = (8 \text{ Mio €} / 6 \text{ Mio €}) * 33,350 \text{ €}$$

$$\text{NNC} = 44,467 \text{ €}$$

Answer: Rounded to the full Euro, the new nominal capital is **NNC = 44,467 €**

4. What are the (rounded) nominal shares?

Answer:

After 2nd round at t2

Partner	Nominal Capital [€]	Actual Shares [%]	Previous Shares [%]	Comment
Founders	25,000	56.22	75	Dilution
Side Investor	3,350	7.53	10	Dilution
HTGF	6,650	14.95	15	Conversion of loan
New Investor	9,500	21.36	--	
Sum	44,467	100,06	100	

REFERENCES

References

www.high-tech-gruenderfonds.de

Brandenburger, M. Frank, G. , Terzidis, O., Presse, A. (2012): *Business Angels Decision Criteria in New Ventures in the Seed and Early Stage Phase*, in: Gossel, Britta, M. (Hrsg.), *Entrepreneurship Research – Discussing Today the Awareness of Tomorrow* (YERC 2012), Band 8, Universitätsverlag Ilmenau, pp. 33-56

Bundesministerium für Wirtschaft und Energie (BMWi) (2017a): *GründerZeiten extra: Start-ups: Finanzierung und Wagniskapital*, Berlin, Januar 2017

Bundesministerium für Wirtschaft und Energie (BMWi) (2017b): *GründerZeiten 06: Existenzgründungsfinanzierung*, Berlin, April 2017

Fritsch, Michael (2016): *Entrepreneurship: Theorie, Empirie, Politik*, Springer: Berlin, Heidelberg

Fueglistaller, Urs., Müller, Christoph, Müller, Susan, Volery, Thierry (2016): *Entrepreneurship: Modelle – Umsetzung – Perspektiven, Mit Fallbeispielen aus Deutschland, Österreich und der Schweiz*, 4. Auflage, Springer: Wiesbaden

Hahn, Christopher (Hrsg.) (2014): *Finanzierung und Besteuerung von Start-up-Unternehmen: Praxisbuch für erfolgreiche Gründer*, Springer: Wiesbaden

References

Hof, Robert (2017): *Gründungsfinanzierung: Wirkung des Innovationsgrades auf das Signaling bei der Eigenkapitalfinanzierung*, in: Brettel, Malte, Koch, Lambert T., Kollmann, Tobias, Witt, Petter (Hrsg.), *Entrepreneurship*, Springer: Wiesbaden

Hornberger, Matthias (2012): *Risikokapitalgeber - Perspektiven und Praxis*. KIT Entrepreneurship Talks. KIZOO. Karlsruhe, 14.06.2012.

Insolvenzordnung: <https://www.gesetze-im-internet.de/insol/>

Kast, Daniel; Alshut, Jörg (Hg.) (2004): *Handbuch für junge Unternehmen*. Heidelberg: Verl. Recht und Wirtschaft (BB-Handbuch).

Pott, O., Pott, A. (2015): *Entrepreneurship: Unternehmensgründung, Businessplan und Finanzierung, Rechtsformen und gewerblicher Rechtsschutz*, 2. Auflage, Springer: Berlin, Heidelberg

Silva, Jorge (2004): Venture capitalists' decision-making in small equity markets. A case study using participant observation. In *Venture capital* 6 (2-3), pp. 125–145.

Volkman, Christine K.; Tokarski, Kim Oliver (2006): *Entrepreneurship. Gründung und Wachstum von jungen Unternehmen*. Stuttgart: Lucius & Lucius

Thank You!

